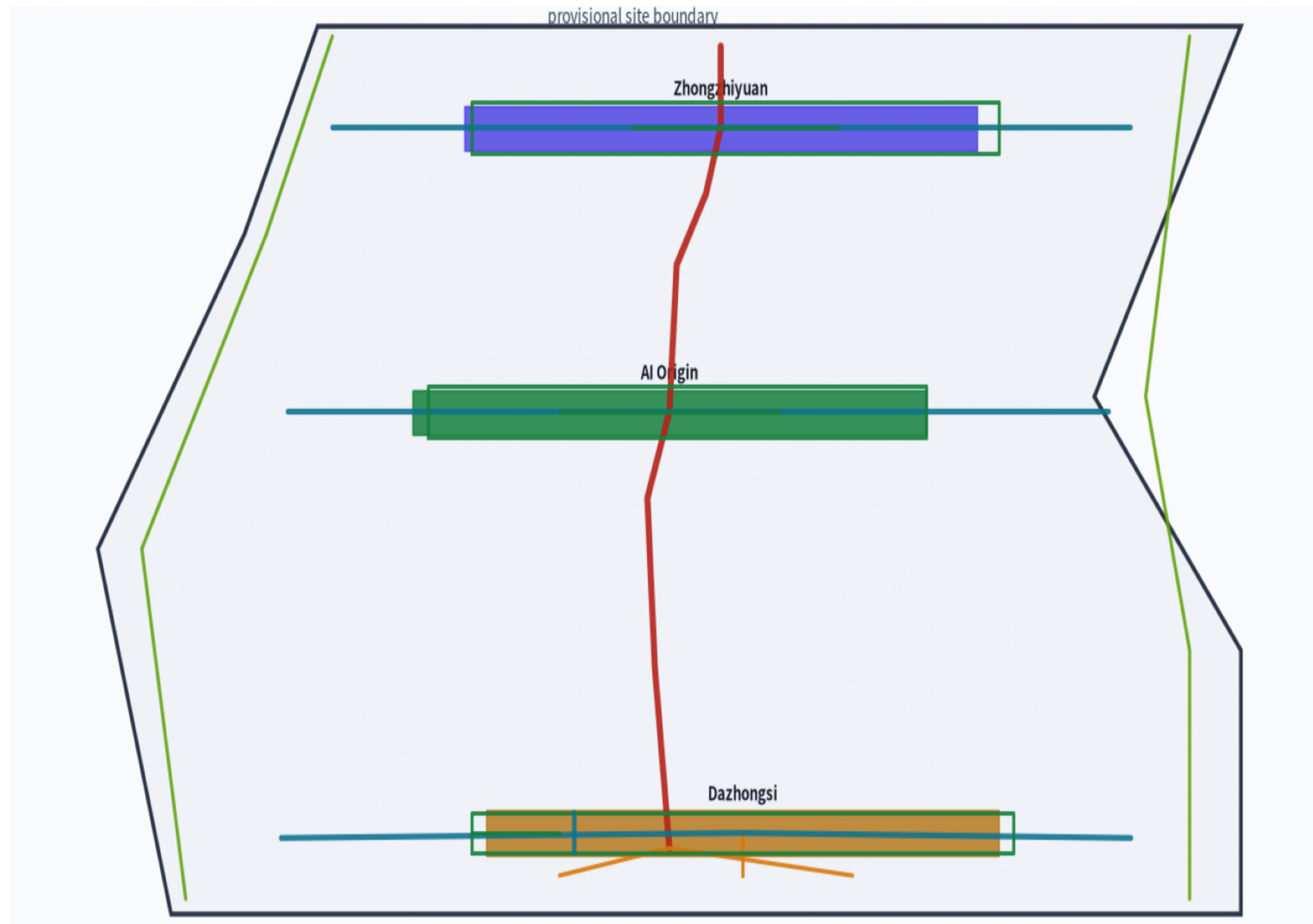


Evidence chain and package



Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective | Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective $f_3 \equiv f_2$ (degenerates to cost; no independent site heritage conclusion)

Three-level scope and spatial structure

Coordinated study scope

43.6 km²

AI ecosystem / three zones two wings / future urban form

Overall design scope

11.4 km²

Urban renewal framework / land use / transport & utilities

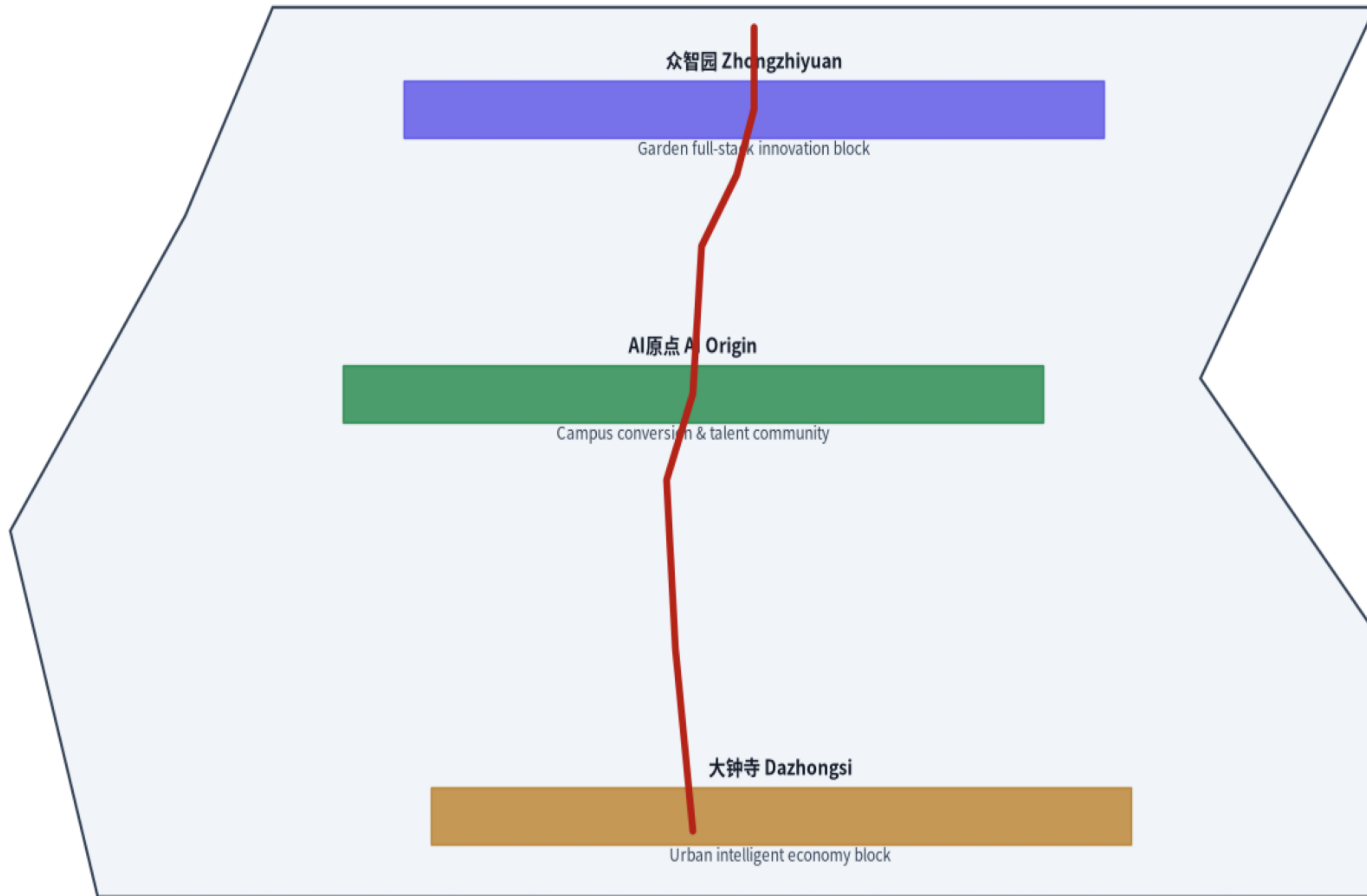
Key area scope

368.4 ha

Three key detailed-design areas

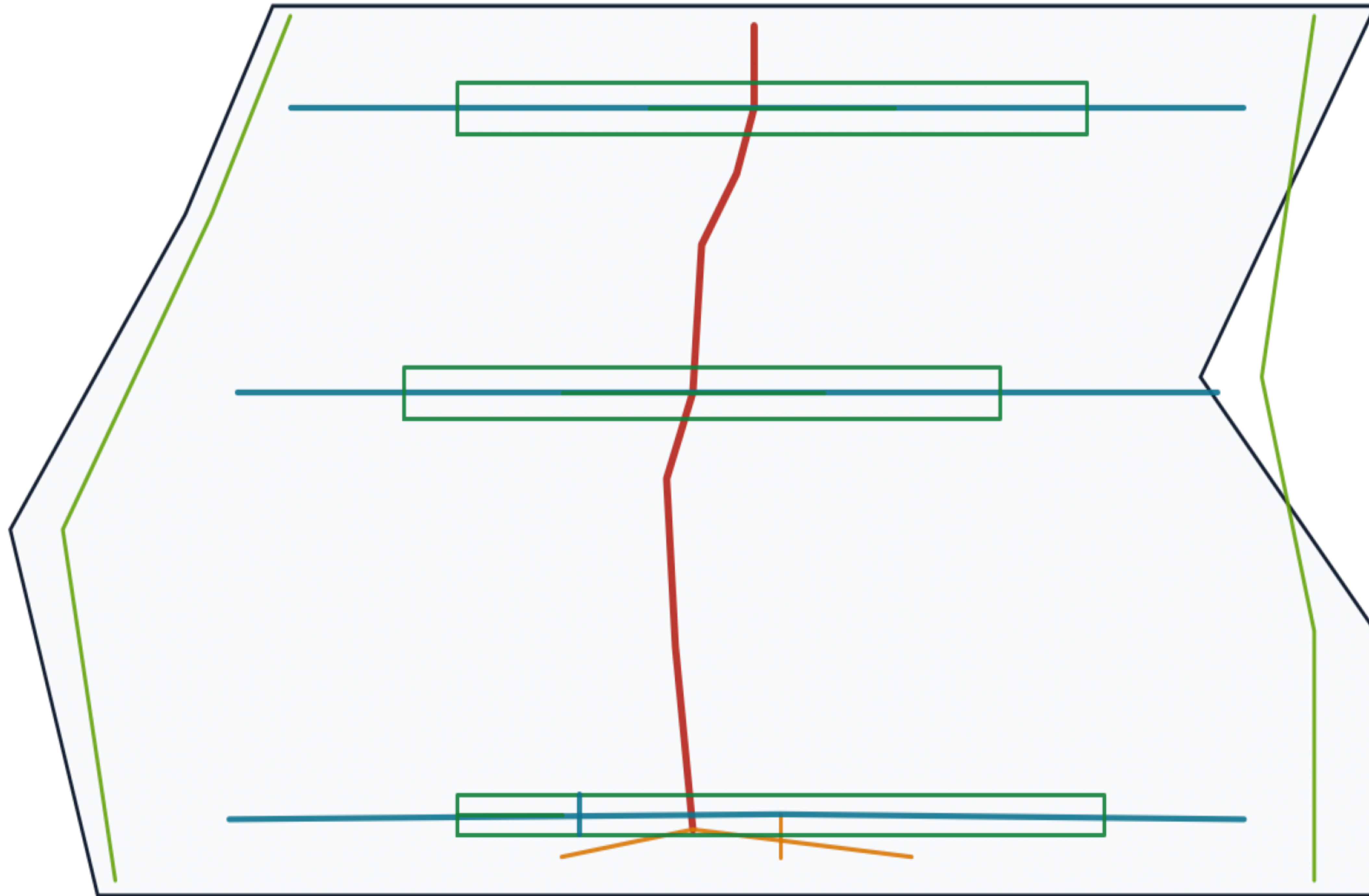
Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Key areas and Physarum spine



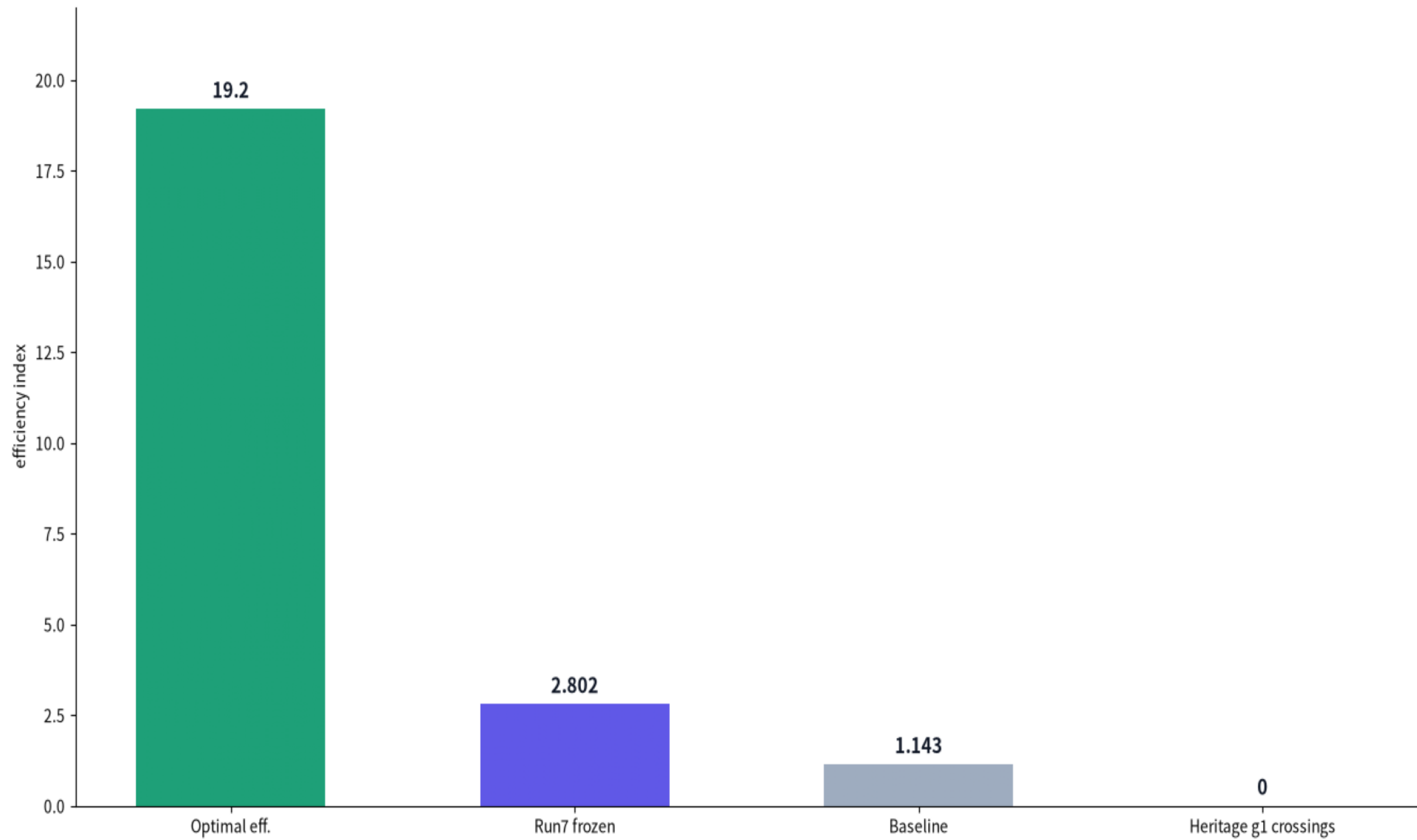
Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Mobility and blue-green public space



Method: Bio-Physarum adaptive network (Tero et al. 2010) + NSGA-II multi-objective

Metrics and method validation



Plan03 urban-integration UDS 80.34 recommended; heritage objective $f_3 \equiv f_2$ degenerates to cost — '0 heritage crossings' is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective $f_3 \equiv f_2$ (degenerates to cost; no independent site heritage conclusion)